contribute to what we observe on the screen, perhaps quantum mechanics is telling us that both potential histories of the electron contribute as well.

Surprisingly, this strange and wonderful idea—the brainchild of the Nobel laureate Richard Feynman, one of the twentieth century's most creative physicists—provides a perfectly viable way of thinking about quantum mechanics. According to Feynman, if there are alternative ways in which a given outcome can be achieved—for instance, an electron hits a point on the detector screen by traveling through the left slit, or hits the same point on the screen but by traveling through the right slit—then there is a sense in which the alternative histories all happen, and happen simultaneously. Feynman showed that each such history would contribute to the probability that their common outcome would be realized, and if these contributions were correctly added together, the result would agree with the total probability predicted by quantum mechanics.

Feynman called this the *sum over histories* approach to quantum mechanics; it shows that a probability wave embodies all possible pasts that could have preceded a given observation, and illustrates well that to succeed where classical physics failed, quantum mechanics had to substantially broaden the framework of history.[3]

To Oz

There is a variation on the double-slit experiment in which the interference between alternative histories is made even more evident because the two routes to the detector screen are more fully separated. It is a little easier to describe the experiment using photons rather than electrons, so we begin with a photon source—a laser—and we fire it toward what is known as a *beam splitter.* This device is made from a half-silvered mirror, like the kind used for surveillance, which reflects half of the light that hits it while allowing the other half to pass through. The initial single light beam is thus split in two, the left beam and the right beam, similar to what happens to a light beam that impinges on the two slits in the double-slit setup. Using judiciously placed fully reflecting mirrors, as in Figure 7.1, the two beams are brought back